

Regression with Panel Data

SW Chapter 10

ECON3500: Econometrics and Applications

Spring 2026

Learning Objectives

By the end of this chapter, you will be able to:

Distinguish **cross-sectional**, **time-series**, and **panel** data

Estimate and interpret **difference-in-differences** regressions

Understand the **first-difference** estimator and why fixed effects drop out

Estimate regressions with **entity** and **time fixed effects**

Explain when **clustered standard errors** are needed

Types of Data

Cross-Sectional Data

A random sample of units observed at **one point in time**.

This is what we've been using so far!

Relationship between hours worked and wages

How CEO tenure relates to compensation

Test scores and class size across California districts

Cross-Sectional Data

id[1] 1											
	id	wage	educ	exper	tenure	nonwhite	female	married	numdep	smsa	northcen
1	1	3.1	11	2	0	0	1	0	2	1	0
2	2	3.2	12	22	2	0	1	1	3	1	0
3	3	3	11	2	0	0	0	0	2	0	0
4	4	6	8	44	28	0	0	1	0	1	0
5	5	5.3	12	7	2	0	0	1	1	0	0
6	6	8.8	16	9	8	0	0	1	0	1	0
7	7	11	18	15	7	0	0	0	0	1	0
8	8	5	12	5	3	0	1	0	0	1	0
9	9	3.6	12	26	4	0	1	0	2	1	0
10	10	18	17	22	21	0	0	1	0	1	0
11	11	6.3	16	8	2	0	1	0	0	1	0
12	12	8.1	13	3	0	0	1	0	0	1	0
13	13	8.8	12	15	0	0	0	1	2	1	0
14	14	5.5	12	18	3	0	0	0	0	1	0
15	15	22	12	31	15	0	0	1	1	1	0
16	16	17	16	14	0	0	0	1	1	1	0
17	17	7.5	12	10	0	0	1	1	0	1	0
18	18	11	13	16	10	0	1	0	0	1	0
19	19	3.6	12	13	0	0	1	1	3	1	0
20	20	4.5	12	36	6	0	1	1	0	1	0
21	21	6.9	12	11	4	0	1	0	0	1	0
22	22	8.5	12	29	13	0	0	1	3	1	0
23	23	6.3	16	9	9	0	1	0	0	1	0
24	24	.53	12	3	1	0	1	0	0	1	0
25	25	6	11	37	8	1	1	0	0	1	0
26	26	9.6	16	3	3	1	0	1	1	1	0
27	27	7.8	16	11	10	0	0	1	1	1	0

Stata data browser: each row is a different individual, observed once

Time-Series Data

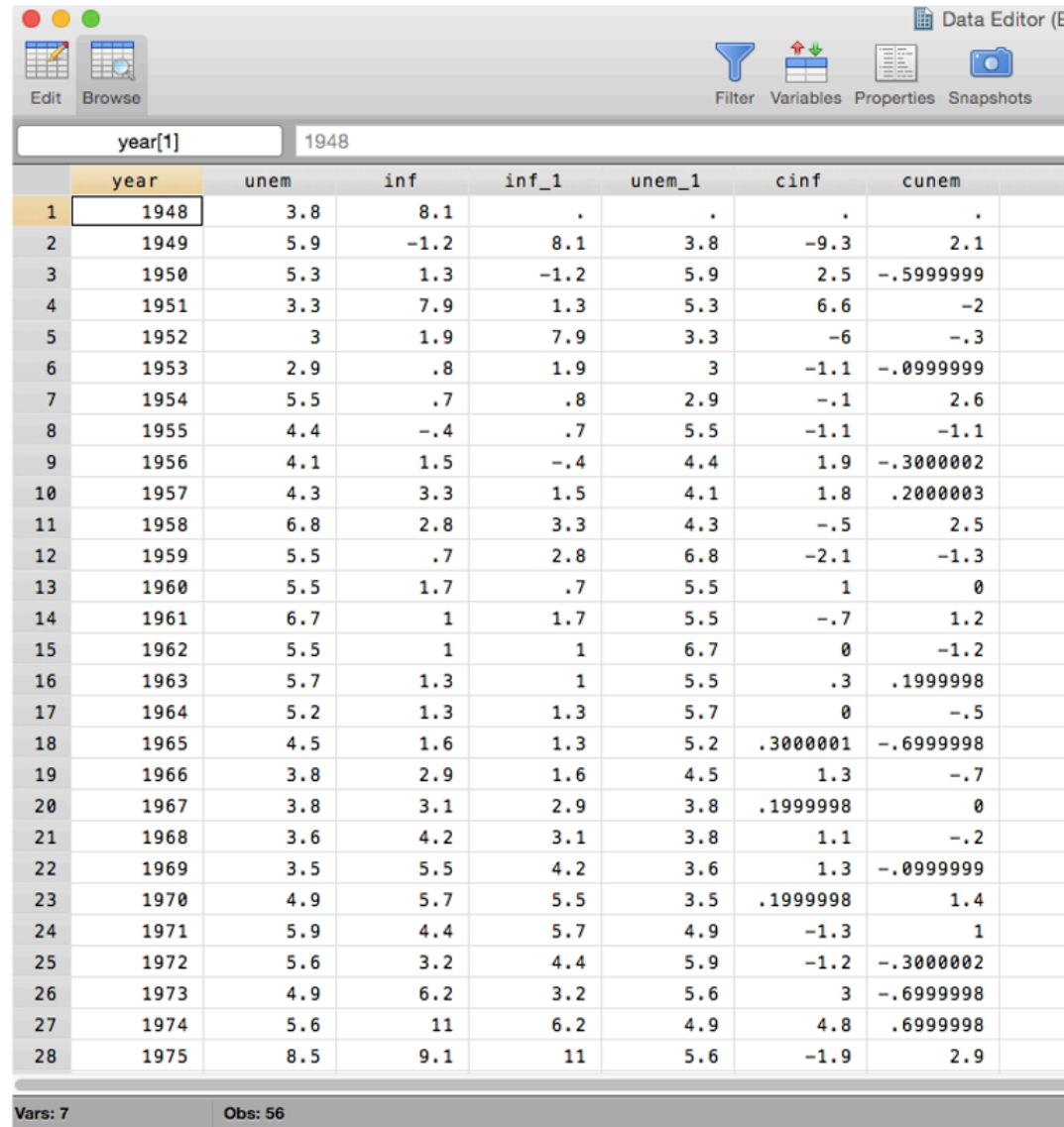
Observations on **one unit over time**.

Stock-market dividends for Apple over the past 10 years

GDP growth over time

Annual infant mortality rates for the U.S.

Time-Series Data



	year	unem	inf	inf_1	unem_1	cinf	cunem
1	1948	3.8	8.1	.	.	.	.
2	1949	5.9	-1.2	8.1	3.8	-9.3	2.1
3	1950	5.3	1.3	-1.2	5.9	2.5	-.5999999
4	1951	3.3	7.9	1.3	5.3	6.6	-2
5	1952	3	1.9	7.9	3.3	-6	-.3
6	1953	2.9	.8	1.9	3	-1.1	-.0999999
7	1954	5.5	.7	.8	2.9	-.1	2.6
8	1955	4.4	-.4	.7	5.5	-1.1	-1.1
9	1956	4.1	1.5	-.4	4.4	1.9	-.3000002
10	1957	4.3	3.3	1.5	4.1	1.8	.2000003
11	1958	6.8	2.8	3.3	4.3	-.5	2.5
12	1959	5.5	.7	2.8	6.8	-2.1	-1.3
13	1960	5.5	1.7	.7	5.5	1	0
14	1961	6.7	1	1.7	5.5	-.7	1.2
15	1962	5.5	1	1	6.7	0	-1.2
16	1963	5.7	1.3	1	5.5	.3	.1999998
17	1964	5.2	1.3	1.3	5.7	0	-.5
18	1965	4.5	1.6	1.3	5.2	.3000001	-.6999998
19	1966	3.8	2.9	1.6	4.5	1.3	-.7
20	1967	3.8	3.1	2.9	3.8	.1999998	0
21	1968	3.6	4.2	3.1	3.8	1.1	-.2
22	1969	3.5	5.5	4.2	3.6	1.3	-.0999999
23	1970	4.9	5.7	5.5	3.5	.1999998	1.4
24	1971	5.9	4.4	5.7	4.9	-1.3	1
25	1972	5.6	3.2	4.4	5.9	-1.2	-.3000002
26	1973	4.9	6.2	3.2	5.6	3	-.6999998
27	1974	5.6	11	6.2	4.9	4.8	.6999998
28	1975	8.5	9.1	11	5.6	-1.9	2.9

Vars: 7 Obs: 56

Stata data browser: each row is a different year for one country

Panel (Longitudinal) Data

Multiple units observed across **multiple time periods** — cross-sections over time.

Track how individuals' earnings change over time

Crime rates by city over time

Very useful for policy analysis!

Panel vs. Pooled Cross-Sections

A **panel** tracks the **same units** over time (Bob in 1990 and Bob in 1991).

A **pooled cross-section** draws **different units** each period (Bob in 1990, Jane in 1991) — like multiple waves of the ACS.

Panel Data

Data Editor (Browse) - PRISON.DTA

Filter Variables Properties Snapshots

state[1] 1

	state	year	govelec	black	metro	unem	criv	crip	lcriv	lcrip
1	1	80	0	.256	.632	.08775	4.447868	44.47638	1.492425	3.794958
2	1	81	0	.2557	.6362	.10667	4.700944	44.24879	1.547763	3.789828
3	1	82	1	.2554	.6404	.14367	4.49758	42.05045	1.503539	3.73887
4	1	83	0	.2551	.6446	.13667	4.186833	37.08439	1.431945	3.613196
5	1	84	0	.2548	.6488	.11167	4.353239	35.04226	1.47092	3.556555
6	1	85	0	.2545	.6530001	.08908	4.630758	35.2668	1.532721	3.562942
7	1	86	1	.2542	.6572	.09833	5.665331	37.8735	1.734365	3.634252
8	1	87	0	.2539	.6614	.07775	5.686924	39.58107	1.73817	3.678351
9	1	88	0	.2536	.6656	.07208	5.728628	41.05592	1.745476	3.714935
10	1	89	0	.2533	.6698	.07025	6.036973	41.25161	1.797903	3.71969
11	1	90	1	.253	.674	.06775	7.069136	41.96889	1.955738	3.736928
12	1	91	0	.2527	.6782	.072	8.439609	45.20342	2.132936	3.811173
13	1	92	0	.2524	.6824	.073	8.712421	43.94321	2.16475	3.782898
14	1	93	0	.2521	.6866	.075	7.804156	40.98352	2.054657	3.71317
15	2	80	0	.034	.434	.09592	4.773632	57.0398	1.563107	4.043749
16	2	81	0	.0347	.4317	.09233	6.069378	58.93301	1.803256	4.076401
17	2	82	1	.0354	.4294	.09983	6.071111	54.39778	1.803542	3.996323
18	2	83	0	.0361	.4271	.10308	6.02459	53.05123	1.795849	3.971258
19	2	84	0	.0368	.4248	.1015	6.046692	53.43969	1.799511	3.978554
20	2	85	0	.0375	.4225	.09608	5.697369	51.85714	1.740004	3.948493
21	2	86	1	.0382	.4202	.10908	5.599265	55.7114	1.722635	4.020185
22	2	87	0	.0389	.4179	.108	4.435993	47.94249	1.489751	3.870002
23	2	88	0	.0396	.4156	.09225	4.948339	41.63469	1.599052	3.728934
24	2	89	0	.0403	.4133	.06742	4.795247	41.25594	1.567625	3.719795
25	2	90	1	.041	.411	.07025	5.207581	45.95126	1.650115	3.827581
26	2	91	0	.0417	.4087	.085	6.149385	50.96661	1.816352	3.931171
27	2	92	0	.0424	.4064	.091	6.593537	49.0068	1.88609	3.891959
28	2	93	0	.0431	.4041	.076	7.607679	48.07178	2.029158	3.872695

Vars: 45 Obs: 714

Stata data browser: same entities tracked across time periods

Panel vs. Repeated Cross-Sections

Repeated (pooled) cross-sections:

Draw randomly from a large population at various points in time

Observations are **independent** over time

Independence $\Rightarrow$ inference is straightforward

Panel data:

Track the **same** observational units over time

Population may be independent at first draw: Bob in 1990 is independent of Jane in 1990

But **dependence over time**: Bob in 1990 is related to Bob in 1991

Detroit will probably have high crime rates next year too

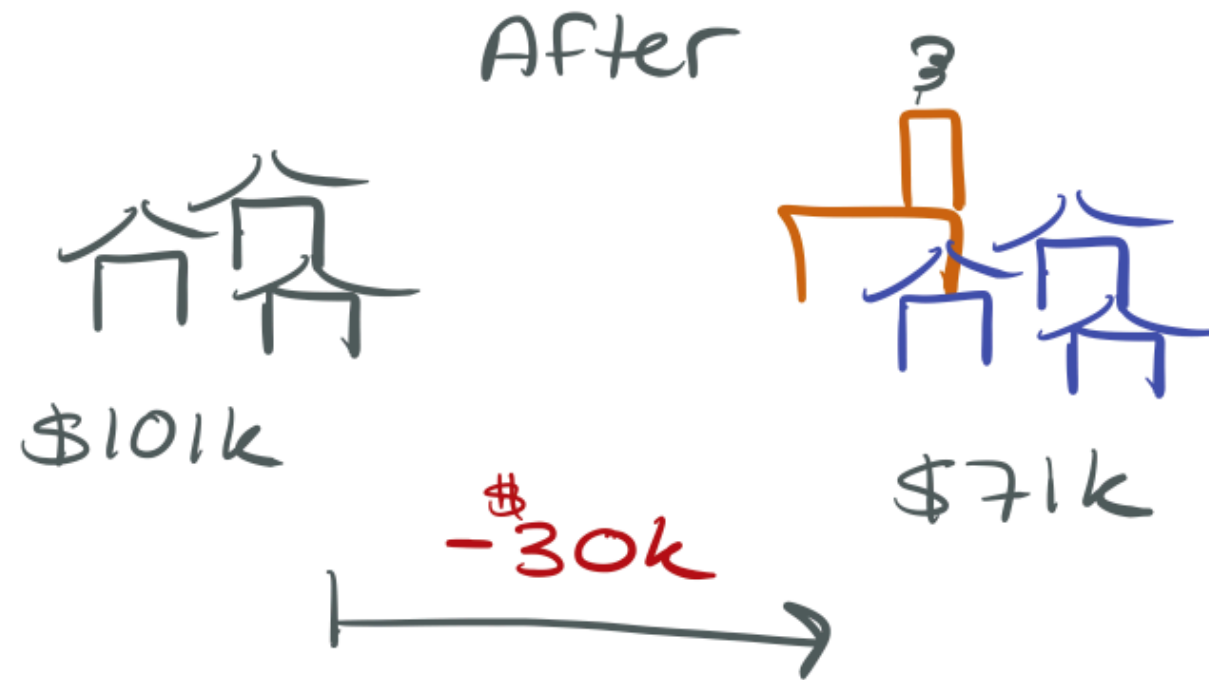
This dependence has important implications for **standard errors** (more on this later).

Difference-in-Differences

Motivating Example: Garbage Incinerator

Question: What is the effect of a garbage incinerator on nearby housing prices?

After the incinerator was built:

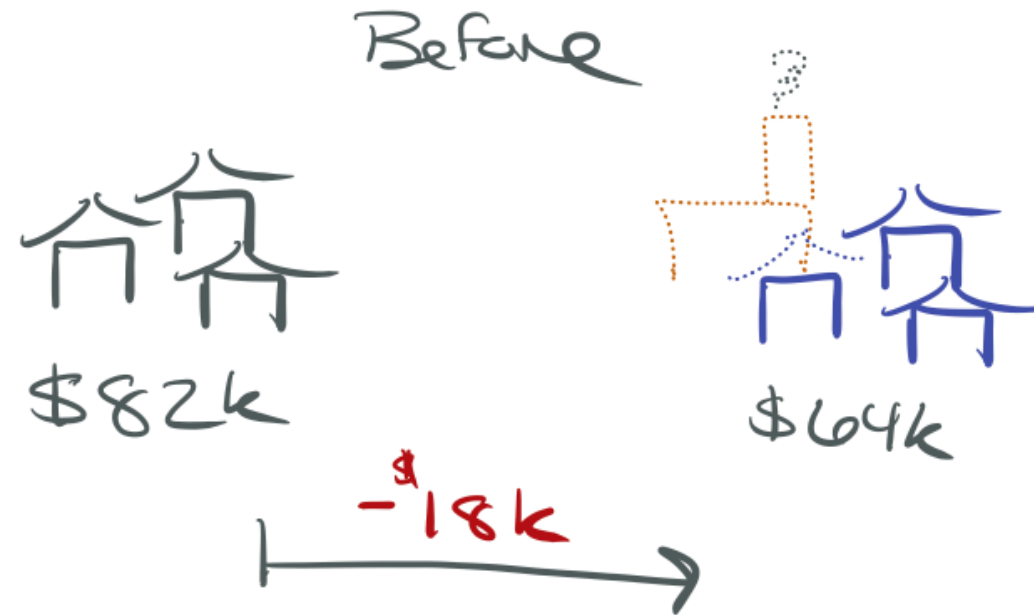


$$\widehat{rprice} = 101,308 - 30,688 \cdot nearinc$$

Houses near the incinerator sell for ~\$30k less. Did the incinerator cause this?

Not So Fast...

Look at the relationship **before** the incinerator was built:

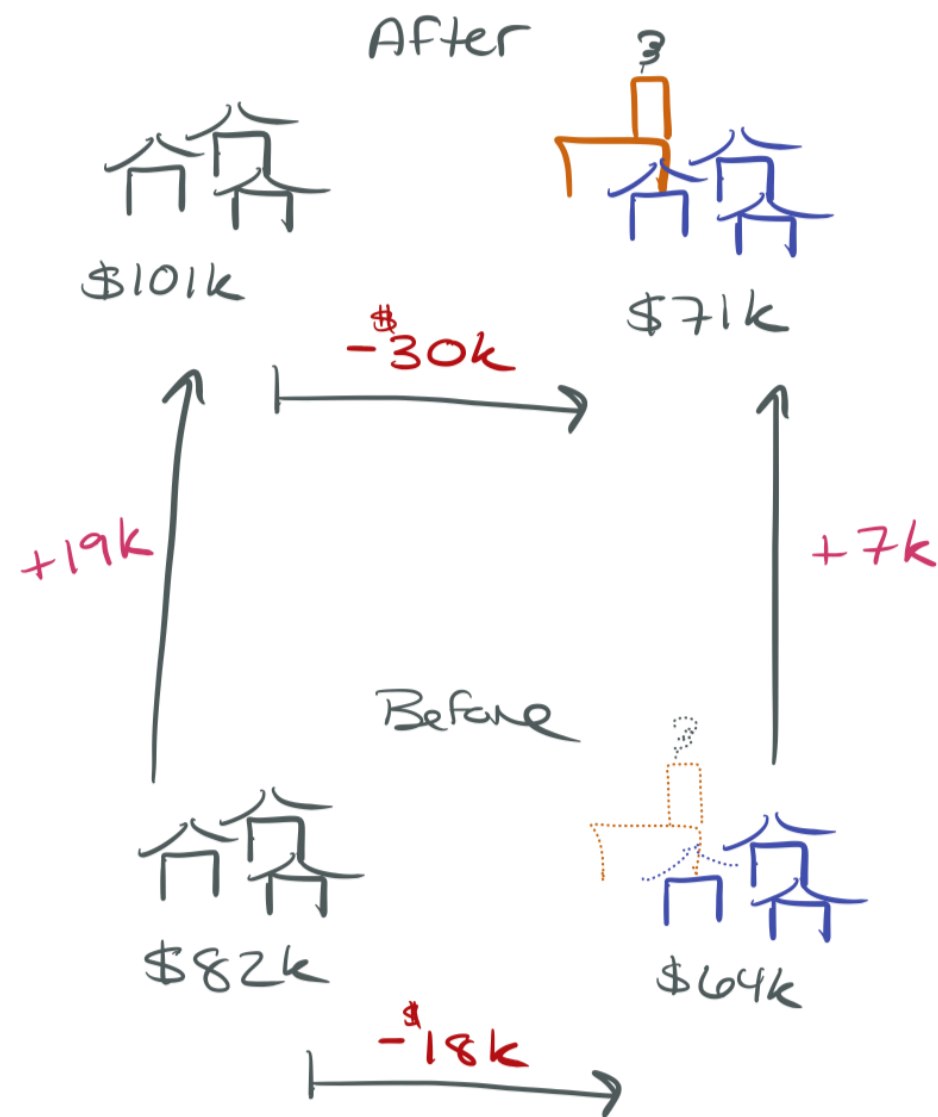


$$\widehat{rprice} = 82,517 - 18,824 \cdot nearinc$$

The incinerator was built in a place where housing prices were **already depressed!**

The \$30k gap reflects both the incinerator effect **and** pre-existing differences.

The Before-and-After Picture



The Difference-in-Differences Estimate

We need to **subtract out** the pre-existing difference:

$$\hat{\delta}_1 = \underbrace{(-30,688)}_{\text{after gap}} - \underbrace{(-18,824)}_{\text{before gap}} = -11,864$$

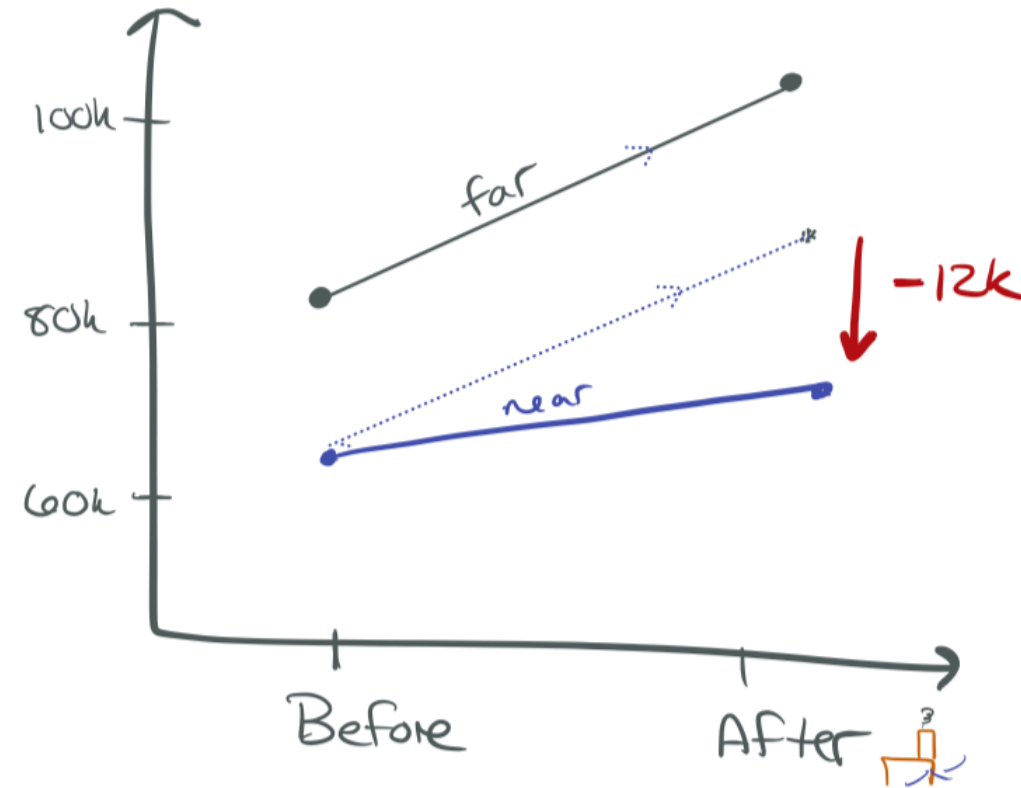
The incinerator reduced nearby prices by ~\$12k — not \$30k.

Equivalently:

$$\hat{\delta}_1 = (\bar{y}_{after,near} - \bar{y}_{after,far}) - (\bar{y}_{before,near} - \bar{y}_{before,far})$$

This is the **difference-in-differences** (DiD) estimator.

DiD Graphically



The DiD estimate is the gap between actual and counterfactual

The dashed line shows where near-incinerator prices **would have been** if they followed the same trend as far-away prices.

DiD in a Regression Framework

We can capture all of this in a single regression:

$$rprice_i = \beta_0 + \delta_0 \cdot after_i + \beta_1 \cdot nearinc_i + \delta_1 \cdot (after_i \times nearinc_i) + u_i$$

	Before ()	After ()	After Before
Control (far)	β_0	$\beta_0 + \delta_0$	δ_0
Treatment (near)	$\beta_0 + \beta_1$	$\beta_0 + \delta_0 + \beta_1 + \delta_1$	$\delta_0 + \delta_1$
Treat – Control	β_1	$\beta_1 + \delta_1$	δ_1

δ_1 is the **DiD estimate** — the coefficient on the interaction term.

Policy Evaluation with DiD

This logic applies broadly to any setting with:

A **before/after** period

Treatment/control groups

$$y_{it} = \beta_0 + \delta_0 \cdot after_t + \beta_1 \cdot treated_i + \delta_1 \cdot (after_t \times treated_i) + u_{it}$$

Adding controls is fine: Additional covariates can be included without changing how δ_1 is identified — they **reduce residual variance** and **improve precision** (tighter standard errors).

! The Parallel Trends Assumption (*common trends assumption in SW*)

DiD requires that in the **absence of treatment**, the gap between treatment and control groups would have stayed **parallel** over time.

There **can** be unobserved factors at one point in time, if they affect both groups equally

The two groups **can** be very different from each other, so long as they are on the same trajectories

💡 Knowledge Check

In the incinerator example, what would violate parallel trends?

...

If, say, a new highway was built near the incinerator site at the same time — affecting near-incinerator prices but not far-away prices — that would break parallel trends.

Two-Period Panel Data: First Differencing

From DiD to Panel Data

Now assume we observe the **same entities** in two time periods.

Example: Unemployment and crime across cities in 1982 and 1987.

Detroit has high unemployment and high crime

Does high unemployment **cause** high crime, or is there another explanation?

Explanatory variables might help — but what if we don't have them?

Key Insight

By observing the same cities in two periods, we can control for **all time-invariant city-specific factors** — even unobserved ones — by examining how outcomes changed *within* each city over time.

The Panel Data Model

$$crmrte_{it} = \beta_0 + \delta_0 \cdot d87_t + \beta_1 \cdot unemp_{it} + \alpha_i + u_{it}$$

where $t = 1982, 1987$

$d87_t$: time dummy for the second period

α_i : **unobserved** time-constant factors (**entity fixed effects**)

- Detroit's culture, geography, historical factors, etc.

u_{it} : other unobserved factors (**idiosyncratic error**)

The problem: α_i is likely correlated with $unemp_{it}$ — cities with certain unobserved characteristics have both high unemployment and high crime. This is **OVB**.

The First-Difference Trick

Write the model for each period:

$$crrmrte_{i,1987} = \beta_0 + \delta_0 + \beta_1 \cdot unemp_{i,1987} + \alpha_i + u_{i,1987}$$

$$crrmrte_{i,1982} = \beta_0 + \beta_1 \cdot unemp_{i,1982} + \alpha_i + u_{i,1982}$$

Subtract the second from the first:

$$\Delta crrmrte_i = \delta_0 + \beta_1 \cdot \Delta unemp_i + \Delta u_i$$

! The Fixed Effect Drops Out!

Because $\alpha_i - \alpha_i = 0$, **all time-invariant unobserved factors** are eliminated. We don't need to measure or control for them.

First-Difference Estimate

Estimate the first-differenced equation by OLS:

$$\Delta \widehat{crime} = 15.4 + 2.22 \cdot \Delta unemp$$

Interpretation: A 1 percentage point increase in the unemployment rate is associated with 2.22 more crimes per 1,000 people.

What First-Differencing Can and Cannot Do

Solves: Time-invariant omitted variable bias (city culture, geography, etc.)

Does not solve: Time-varying endogeneity (factors that change across cities over time)

Limitation: Imprecise if X varies little over time; impossible to estimate effects of time-invariant variables

Fixed Effects Estimation

The Idea Behind Fixed Effects

Instead of differencing, what if we **control for** each entity directly?

Consider a panel of wages over time. There might be unobserved Bob-specific characteristics:

Bob is hard-working (+)

Bob is a competitive negotiator (+)

Bob smells funny (–)

We can never control for all of Bob's unique quirks.

But we can **“control for” Bob** — give Bob his very own **fixed effect**: the unique “stuff” that Bob brings to the table.

This only covers things that **stay constant over time**.

Fixed Effects: The Math

Start with:

$$y_{it} = \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + \alpha_i + u_{it}$$

Average over time for each entity:

$$\bar{y}_i = \beta_1 \bar{x}_{i1} + \cdots + \beta_k \bar{x}_{ik} + \alpha_i + \bar{u}_i$$

Subtract (time-demeaning):

$$[y_{it} - \bar{y}_i] = \beta_1 [x_{it1} - \bar{x}_{i1}] + \cdots + \beta_k [x_{itk} - \bar{x}_{ik}] + [u_{it} - \bar{u}_i]$$

Because $\alpha_i - \alpha_i = 0$, the fixed effect drops out!

Estimate the time-demeaned equation by OLS

This uses **within-entity variation** only: how X and Y move **within** the same entity over time

This is called **within estimation**

Three Ways to Estimate FE in Stata

Option 1: Manual Demeaning

Demean your data by hand, then estimate OLS on demeaned data.

Option 2: Dummy Variables / `areg`

Include entity dummies directly, or use `areg` which “absorbs” one set of fixed effects:

```
areg scrap grant d88 d89, absorb(fcode) cluster(fcode)
```

Option 3: `xtreg` (Recommended)

Tell Stata you have panel data, then estimate with the fixed effects option:

```
xtset fcode year  
xtreg scrap grant d88 d89, fe vce(cluster fcode)
```

Entity Fixed Effects: What They Control For

! Key Principle

Entity fixed effects control for **all time-invariant factors** — observed or unobserved.

You don't need to include time-invariant covariates in a fixed-effects model — they are already accounted for!

Practical notes:

We don't usually interpret the fixed effect coefficients themselves

Which entity is the “omitted” group (avoiding the dummy variable trap) doesn't matter — we're not interpreting them

Fixed effects are typically not reported in regression output

Time Fixed Effects

What about shocks that affect **all entities equally** at a given time?

A national recession hits all cities in 2008

A federal policy change affects all states simultaneously

Inflation affects all firms' costs in the same year

Add **time fixed effects**: a dummy for each time period.

$$y_{it} = \beta_1 x_{it} + \alpha_i + \lambda_t + u_{it}$$

α_i : entity fixed effect (absorbs entity-specific, time-invariant factors)

λ_t : time fixed effect (absorbs time-specific, entity-invariant factors)

Entity + Time Fixed Effects: What's Left?

With **both** entity and time fixed effects:

$$y_{it} = \beta_1 x_{it} + \alpha_i + \lambda_t + u_{it}$$

Controlled for:

All time-invariant entity characteristics (α_i)

All entity-invariant time shocks (λ_t)

Not controlled for:

Factors that vary **across both entities and time**

Knowledge Check

You estimate the effect of a state-level minimum wage increase on employment, with state and year fixed effects. What kind of omitted variable could still bias your results?

...

A factor that varies across states *and* over time — e.g., state-specific economic booms that coincide with minimum wage increases.

Assumptions and Inference

Least Squares Assumptions for Panel Data

Consider a single X :

$$Y_{it} = \beta_1 X_{it} + \alpha_i + u_{it}, \quad i = 1, \dots, n; \quad t = 1, \dots, T$$

$$E[u_{it} | X_{i1}, \dots, X_{iT}, \alpha_i] = 0$$

- u_{it} cannot be correlated with **any present, past, or future** values of X !

$(X_{i1}, \dots, X_{iT}, u_{i1}, \dots, u_{iT})$ are i.i.d. draws across entities

- Only need independence **across** entities, not within

(X_{it}, u_{it}) have finite fourth moments

No perfect multicollinearity

If these hold, FE estimates are **consistent and normally distributed** for large n .

The Problem of Serial Correlation

$$Y_{it} = \alpha_i + \beta_1 X_{it} + \omega_{it}$$

The time-varying error ω_{it} reflects factors that change over time but affect the outcome.

Example: If Y_{it} is traffic fatalities, maybe ω_{it} includes road quality in state i .

Road quality last year $\approx$ road quality this year

$\omega_{i,t}$ and $\omega_{i,t-1}$ are **correlated**

This is **autocorrelation** (or serial correlation) — observations on the same entity over time are not independent.

Why Autocorrelation Matters

Previously, we assumed different observations were independent.

Implausible for data on the same entity over time

Without correction, we estimate **incorrect standard errors**

Confidence intervals will **not have 95% coverage**

⚠ Autocorrelation Does Not Bias $\hat{\beta}$

The coefficients are still consistent — but our **inference** (standard errors, p -values, CIs) is wrong.

The Solution: Clustered Standard Errors

Clustered standard errors allow for autocorrelation within entities.

Idea: Allow the error terms ω_{it} for different time periods **within the same entity** to be correlated, while maintaining independence **across** entities.

In Stata:

```
* With regress or areg:  
regress y x i.entity, cluster(entity)  
areg y x, absorb(entity) cluster(entity)  
  
* With xtreg:  
xtreg y x, fe vce(cluster entity)
```

Always Cluster in Panel Data

If you're using panel data, you should **almost always** cluster your standard errors at the entity level. Failing to do so overstates precision and leads to false rejections.

Recent Developments in DiD

The Staggered Adoption Problem

The classic DiD has two groups and two periods. In practice, policies are often adopted by **different units at different times** — **staggered adoption**.

Example: States adopt a job training program in different years.

The natural approach: two-way fixed effects (TWFE).

$$y_{it} = \beta_1 \cdot treated_{it} + \alpha_i + \lambda_t + u_{it}$$

Recent research (roughly 2018–2024) has shown that this “obvious” approach can produce **severely misleading estimates** — including **wrong signs**.

Let’s see exactly how this happens.

A Concrete Example: Setup

Suppose a job training program has a **true effect** that **grows over time**:

Year 1 after adoption: productivity rises by **+10**

Year 2 after adoption: productivity rises by **+50** (the program grows dramatically)

Two states, three years. Without treatment, both would follow the same trend:

	2018	2019	2020
No-treatment baseline (both states)	100	105	110

State E (Early): adopts the program in **2019**

State L (Late): adopts the program in **2020**

No pure control group exists (both eventually get treated)

What Actually Happens

	2018	2019	2020
State E (treats in 2019)	100	115	160
State L (treats in 2020)	100	105	120

State E in 2019: $105 + 10 = 115$ (year 1 effect)

State E in 2020: $110 + 50 = 160$ (year 2 effect — program grows dramatically)

State L in 2020: $110 + 10 = 120$ (year 1 effect)

The true effect for State L in 2020 is **+10**. Can TWFE recover this?

What TWFE Does

With no pure control group, TWFE takes a **weighted average** of all possible 2×2 DiD comparisons — including contaminated ones.

Comparison 1 — valid: E treated vs. L as untreated control (2018→2019)

$$\underbrace{(115 - 100)}_{\Delta \text{State E}} - \underbrace{(105 - 100)}_{\Delta \text{State L}} = 15 - 5 = [+10]. \textit{kw}$$

Comparison 2 — contaminated: L treated vs. E **already-treated** as “control” (2019→2020)

$$\underbrace{(120 - 105)}_{\Delta \text{State L}} - \underbrace{(160 - 115)}_{\Delta \text{State E}} = 15 - 45 = [-30]. \textit{alert}$$

With equal-sized groups, TWFE weights these comparisons equally:

$$\hat{\beta}_{TWFE} = \frac{1}{2}(+10) + \frac{1}{2}(-30) = [-10]. \textit{alert}$$

❗ TWFE Says the Effect is -10

All true effects are **positive** (+10 or +50), but TWFE estimates **-10**. The sign flipped!

Why? State E’s outcome rose by 45 from 2019→2020 — not from a trend, but because its **treatment effect grew** from +10 to +50. TWFE treats this jump as a “normal trend” in the control, making State L’s +15 gain look like a relative **decline**.

The Core Problem

Goodman-Bacon (2021) showed this formally: the TWFE estimate is a **weighted average** of all possible 2×2 DiD comparisons.

Some comparisons are fine:

Newly treated vs. never treated — exactly what we want

Some are problematic:

Newly treated vs. already treated — uses treated units as “controls”

When treatment effects **change over time** (grow, fade, or vary across groups), the already-treated comparisons introduce bias — and can even get **negative weights**.

This is not a rare edge case. Dynamic treatment effects are the norm in applied economics: job training programs build skills gradually, minimum wage effects accumulate, policy impacts evolve.

New Estimators

Several estimators have been developed to handle staggered adoption correctly:

Callaway and Sant'Anna (2021): Estimate group-time-specific effects, then aggregate. Only uses not-yet-treated or never-treated units as controls.

Sun and Abraham (2021): Interaction-weighted estimator that corrects for heterogeneous effects across adoption cohorts.

de Chaisemartin and D'Haultfoeuille (2020): Identifies which comparisons get negative weights and proposes robust alternatives.

All share one insight: **never use already-treated units as controls.**

What This Means for Practice

Practical Guidance

Classic 2x2 DiD (two groups, two periods): TWFE is **fine**. The new concerns don't apply.

Staggered adoption with **constant treatment effects**: TWFE is usually okay.

Staggered adoption with **dynamic or heterogeneous effects**: Use a robust estimator.

How to check: Run both TWFE and a robust estimator. If results are similar, the standard approach is probably fine. If they diverge, trust the robust estimator.

Beyond ECON3500

You won't need to implement these in this course. But if you do applied research with staggered DiD (very common in policy evaluation), you should be aware of these issues. Stata packages: `csdid`, `did_multiplgt`, `eventstudyinteract`.

Putting It All Together

Comparison of Panel Data Methods

Method	Data Required	What It Controls	Key Assumption	What Remains
DiD	2 groups, 2 periods	Group-level differences; common time trends	Parallel trends	Group-specific time-varying shocks
First Difference	Panel (same units, 2+ periods)	All time-invariant entity factors	No time-varying OVB	Time-varying omitted variables
Entity FE	Panel (same units, 2+ periods)	All time-invariant entity factors	No time-varying OVB	Time-varying omitted variables
Entity + Time FE	Panel (same units, 2+ periods)	Time-invariant entity factors + entity-invariant time shocks	No entity- and time-varying OVB	Entity-time-specific shocks

When to Use What

Difference-in-Differences:

Repeated cross-sections or panel data

Clear treatment/control groups and before/after periods

Need parallel trends assumption

First Differencing:

Panel data with 2 periods

Equivalent to entity FE with 2 periods

Simple, transparent

Fixed Effects:

Panel data with 2+ periods

More efficient than first differencing with $T > 2$ (when errors are serially uncorrelated)

Can include entity FE, time FE, or both

Knowledge Check

A researcher has data on 50 states over 20 years and wants to estimate the effect of a policy that was adopted by some states at different times. Which method is most appropriate?

...

Entity + time fixed effects — controls for permanent state differences and common national trends. This is sometimes called a “generalized DiD” or “two-way fixed effects” design.

Common Pitfalls

1. Forgetting to Cluster

Panel data almost always requires clustered standard errors. Unclustered SEs will be **too small**.

2. Trying to Estimate Time-Invariant Effects

With entity FE, you **cannot** estimate the effect of variables that don't change over time (e.g., gender, race in a person-panel). The fixed effect already absorbs them.

3. Assuming FE Solves All OVB

Fixed effects only eliminate **time-invariant** omitted variables. Time-varying confounders can still bias your estimates.

Key Takeaways

Panel data tracks the same entities over time — unlike pooled cross-sections

DiD compares changes over time across treatment and control groups; requires **parallel trends**

First differencing and **fixed effects** eliminate all time-invariant omitted variables by exploiting **within-entity variation**

With entity + time FE, only **entity-and-time-varying** confounders remain a threat

Always use **clustered standard errors** with panel data to account for autocorrelation

Fixed effects **cannot estimate** effects of time-invariant variables — that's the price of eliminating time-invariant OVB